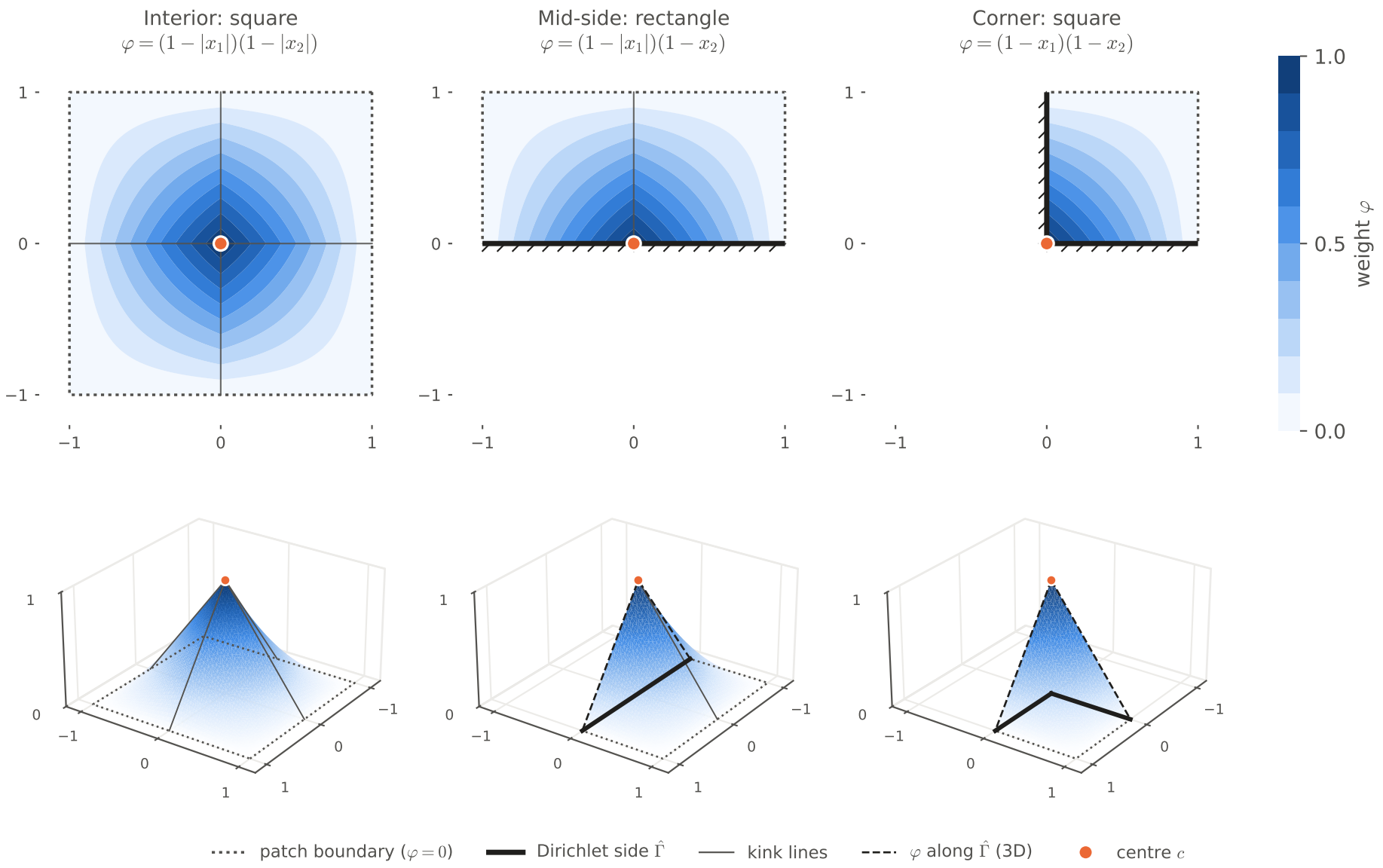

Option 2: bilinear hats on the square patches ($S \equiv 1$)
Interior: square
$\varphi = (1 - |x_1|)(1 - |x_2|)$
Mid-side: rectangle
$\varphi = (1 - |x_1|)(1 - x_2)$
Corner: square
$\varphi = (1 - x_1)(1 - x_2)$
weight φ
1.0
0.5
0.0
patch boundary ($\varphi = 0$)
Dirichlet side $\hat{\Gamma}$
kink lines
φ along $\hat{\Gamma}$ (3D)
centre c